

# MAT 4233 — Weekly Schedule

Modern Abstract Algebra · Fall 2026 · UT San Antonio

## Texts

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### Primary text

- **KvP** — Kramer & von Pippich, *From Natural Numbers to Quaternions*
- **Beardon** — Beardon, *Algebra and Geometry*

### Supplementary reading

- **Needham** — Needham, *Visual Complex Analysis, 25th Anniversary Edition*

### Handouts & papers

- **Conrad** — Conrad, *The Gaussian Integers*
- **Arnold–Rogness** — Arnold & Rogness, *Möbius Transformations Revealed*

## Schedule

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| Date       | Topic                                                                                 | Reading                                               |
|------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|
| Thu 20 Aug | Lecture — Course launch; divisibility and the division algorithm                      | <a href="#">KvP §1.1-1.2</a>                          |
| Tue 25 Aug | Lecture — Primes, the Fundamental Theorem of Arithmetic, gcd and lcm                  | <a href="#">KvP §1.3-1.5</a>                          |
| Thu 27 Aug | Lecture — Semigroups, monoids, groups — all of them $\mathbb{Z}/n\mathbb{Z}$          | <a href="#">KvP §2.1</a>                              |
| Tue 1 Sep  | Lecture — Subgroups; the unit group $(\mathbb{Z}/n\mathbb{Z})^*$ ( <b>Due:</b> Set 1) | <a href="#">KvP §2.2</a>                              |
| Thu 3 Sep  | <b>Q&amp;A</b>                                                                        |                                                       |
| Tue 8 Sep  | Lecture — Group homomorphisms; the order of an element                                | <a href="#">KvP §2.3</a>                              |
| Thu 10 Sep | Lecture — Cyclic groups; primitive roots                                              | <a href="#">KvP §2.3</a>                              |
| Tue 15 Sep | Lecture — Cosets and Lagrange’s theorem ( <b>Due:</b> Set 2)                          | <a href="#">KvP §2.4</a>                              |
| Thu 17 Sep | <b>Q&amp;A</b>                                                                        |                                                       |
| Tue 22 Sep | Lecture — Normal subgroups and quotient groups                                        | <a href="#">KvP §2.5</a>                              |
| Thu 24 Sep | Lecture — The homomorphism theorem                                                    | <a href="#">KvP §2.6</a>                              |
| Tue 29 Sep | Lecture — Euler’s theorem, fast modular exponentiation, and RSA end to end            | <a href="#">KvP App. B</a> ; <a href="#">KvP §2.4</a> |
| Thu 1 Oct  | <b>Exam</b> — Midterm Exam 1 — Segment 1                                              |                                                       |
| Tue 6 Oct  | Lecture — Rings and subrings                                                          | <a href="#">KvP §3.2</a>                              |
| Thu 8 Oct  | Lecture — Ring homomorphisms                                                          | <a href="#">KvP §3.3</a>                              |
| Thu 15 Oct | Lecture — Ideals                                                                      | <a href="#">KvP §3.3</a>                              |
| Tue 20 Oct | Lecture — Quotient rings ( <b>Due:</b> Set 3)                                         | <a href="#">KvP §3.3</a>                              |

| Date       | Topic                                                                                                                                                          | Reading                                                                                                                              |
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| Thu 22 Oct | <b>Q&amp;A</b>                                                                                                                                                 |                                                                                                                                      |
| Tue 27 Oct | Lecture — Fields and skew fields; $F_p$                                                                                                                        | KvP §3.4                                                                                                                             |
| Thu 29 Oct | Lecture — Integral domains and the field of fractions                                                                                                          | KvP §3.5                                                                                                                             |
| Tue 3 Nov  | Lecture — Unique factorization: UFDs ( <b>Due:</b> Set 4)                                                                                                      | KvP §3.7                                                                                                                             |
| Thu 5 Nov  | <b>Q&amp;A</b>                                                                                                                                                 |                                                                                                                                      |
| Tue 10 Nov | Lecture — PIDs and Euclidean domains                                                                                                                           | KvP §3.7                                                                                                                             |
| Thu 12 Nov | Lecture — Complex numbers: arithmetic, the plane, polar form, roots of unity                                                                                   | Beardon §3.1-3.2, §3.5; KvP §5.1-5.2                                                                                                 |
| Tue 17 Nov | Lecture — Mobius transformations: the formula, composition as matrix multiplication, circles to circles, the Riemann sphere, fixed points ( <b>Due:</b> Set 5) | Beardon Ch. 13; Arnold–Rogness; Needham Ch. 3 §I-II, §IV-VI (handout; a copy is on reserve); Mobius Transformations Revealed (2 min) |
| Thu 19 Nov | <b>Q&amp;A</b>                                                                                                                                                 |                                                                                                                                      |
| Tue 24 Nov | Lecture — The Gaussian integers: norm, division with remainder, irreducibles, and sums of two squares — a survey                                               | Conrad                                                                                                                               |
| Tue 1 Dec  | <i>Review</i> — Review — the exam pool                                                                                                                         |                                                                                                                                      |
| Thu 3 Dec  | <b>Exam</b> — Midterm Exam 2 — Segment 2 weighted                                                                                                              |                                                                                                                                      |